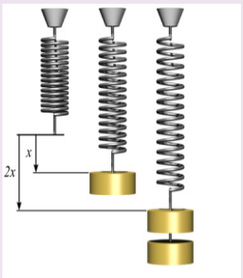
You find that $\frac{1}{200}=\frac{2}{400}=\frac{3}{600}=\frac{4}{800}=\frac{5}{1000}=\ldots$

This kind of proportionate variation is known as Direct variation. Here to find the cost, the weight is multiplied by the constant 200.

If we denote the variable weight as x and the variable cost as y we can express this algebraically as $y = 200x$. The multiplying constant here is 200.

If $\frac{y}{x}=k$ where k is a positive number (a constant), then x and y are said to vary directly. Here, k is known as the constant of proportionality.

Mathematics in real life: This figure shows that doubling the force doubles the displacement. This is a consequence of what is known as *Hooke's law*. It states $F = kx$ where F is the force needed to produce a displacement of x in the position of a spring. To double the displacement, you double the force on the spring; the constant of proportionality k depends on the stiffness of the spring. So this is an example of a direct proportionality.



Visualising Direct variation:

To identify direct variation is to look at the equation and determine if it is of the form $y = kx$, where k is the constant of proportionality. Thus, an equation like $y = 5x$ will always indicate direct proportion among variables.

Thinking Corner

What can you say if the variables x and y are related by the equation $3y - 7x = 0$? It also indicates direct variation. How? Think about it. In that case, what is the constant of proportionality?

Observe this graph:

The distance travelled and the time taken are proportional, but how do we know that?

Note that

i) The graph is a straight line.

ii) The line passes through the origin. When both of these features are present we know that the two quantities on the graph must be directly proportional.

Do you see this in the graph?

Time (in minutes)	4	8	12	16
Distance (in km)	8	16	24	32

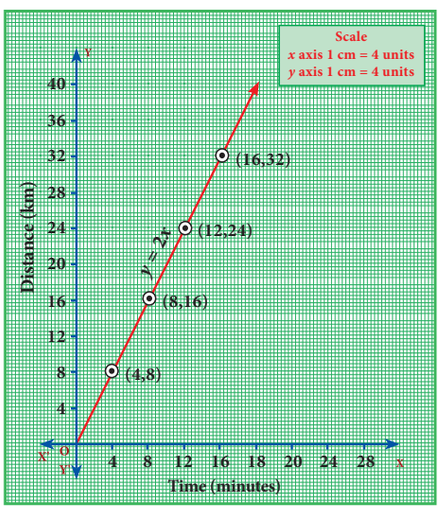


Fig. 3.9

If one variable doubles, the other also doubles. From this you can see the relation $d = rt$ and it is easy to guess the constant of proportionality.

10th_Maths_Chapter 3_English.indd 124 12/23/2024 5:33:53 PM